Extension Note: Shared-Phase Wave Pairs, Malus's Law, and Why the First Detector Lock Must Entrain the Shared Phase

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Companion Note to "A Field–Matter Selector for Outcome Production via Conserving Complex-Interference Exchange

October 14, 2026

1 Purpose of This Note

In standard treatments of quantum entanglement (and in earlier drafts of the selector paper), two-particle correlations are written in **probability / Hilbert-space matrix notation**: a joint coefficient matrix c_{ij} is defined in configuration space, the first detection selects row i with marginal probability $m_i = \sum_j |c_{ij}|^2$, and the second station's probabilities are updated via Bayes' conditional rule $P(j \mid i) = |c_{ij}|^2/m_i$.

While mathematically convenient, that notation obscures the physical wave mechanism by moving the explanation back into probability space. This note records the purely **wave-mechanical formulation in 3D detector space**, answering three foundational questions:

1. How do entangled pair correlations and no-signaling arise from **classical Malus's law** ($\cos^2(a-b)$) and **wave trigonometry** without invoking probability-space matrices c_{ij}?
2. Does this eliminate "spooky action at a distance," or where does the nonlocality reside?
3. Why must the first detector's lock **actively shift (entrain/pin) the shared phase**, rather than the two waves simply carrying a passive, pre-set phase θ_0 from the source?

2 Single-Event Wave Picture of an Entangled Pair (No Probability Space)

Suppose two waves A and B are created together in a single source excitation and travel to two spatially separated polarization analyzers: Polarizer A (oriented at angle a) and Polarizer B (oriented at angle b). Each polarizer splits its incident wave into two orthogonal output ports ($+$ along the polarizer axis, and $-$ along the perpendicular axis $+90°$), where detector sites run the conserving complex-interference walk:

$$\Delta s_+ = -\Delta s_- = 2\epsilon\sqrt{s_+ s_-}\cos(\Delta\phi). \tag{1}$$

Instead of a Hilbert-space probability table c_{ij}, consider what the physical waves do in a single event:

1. **Creation with a Shared, Unpinned Phase Reference:** Because waves A and B originate from the same atomic transition or parametric down-conversion vertex, their

relative phase/polarization difference is locked ($\phi_A - \phi_B = 0$), while their **common absolute orientation is initially unpinned** (free to rotate, like a holographic wavefront prior to fixing the reference beam).

2. **At the First Detector to Lock (Say Station A):** Because the pair's common polarization orientation is unpinned relative to angle a, the two orthogonal output ports of Polarizer A (+ at angle a, and − at angle $a+90°$) receive equal initial wave energy ($s_+(0)_A = 0.5, s_-(0)_A = 0.5$). Station A's local $\sqrt{s_+ s_-}\cos\Delta\phi$ walk runs with its local detector and vacuum phases until one port reaches 1—locking onto either angle a (port +) or angle $a + 90°$ (port −), each occurring in 50% of events.

3. **Entrainment of the Shared Phase Frame:** When Station A's absorber sites lock onto port + (angle a), that lock **physically pins (entrains) the shared polarization orientation of the pair wave to angle** a ($\phi_A = a$). Because the pair wave maintains the rigid phase relation $\phi_A - \phi_B = 0$, the partner wave arriving at Station B now has its polarization oriented along angle a ($\phi_B = a$).

4. **Malus's Law Projection at the Partner Station (Station B):** When a wave polarized along angle a passes through Polarizer B oriented at angle b, **ordinary classical wave optics (Malus's law, 1809)** dictates that the transmitted field amplitude along b is $\cos(a - b)$ and the orthogonal field amplitude along $b + 90°$ is $\sin(a - b)$. By Postulate 1, the initial mechanical oscillator energies deposited in Station B's + and − ports are simply the classical Malus wave intensities:

$$s_+(0)_B = \cos^2(a - b), \qquad s_-(0)_B = \sin^2(a - b). \tag{2}$$

Station B then runs its own local $\sqrt{s_+ s_-}\cos\Delta\phi$ walk on those mechanical shares. By Theorem 1, Station B registers + in a fraction $\cos^2(a-b)$ of those events and − in a fraction $\sin^2(a-b)$.

3 Why "No-Signaling" Is Pure Wave Trigonometry

In this wave-mechanical picture, why does rotating Polarizer A to any angle a never change the single-station click rate at Station B?

- In the 50% **of events** where Station A locks onto its + port (angle a), the partner wave is pinned to angle a, so Station B's + port receives mechanical wave energy $s_+(0)_B^{(+)} = \cos^2(a-b)$.
- In the **other** 50% **of events** where Station A locks onto its − port (angle $a + 90°$), the partner wave is pinned to angle $a + 90°$, so Station B's + port receives mechanical wave energy $s_+(0)_B^{(-)} = \cos^2(a + 90° - b) = \sin^2(a - b)$.
- Averaging over both equally frequent outcomes at Station A, the mean mechanical wave energy delivered to Station B's + port is governed by the elementary trigonometric identity:

$$\bar{s}_+(0)_B = \frac{1}{2}\cos^2(a - b) + \frac{1}{2}\sin^2(a - b) = \frac{1}{2}\left[\cos^2(a - b) + \sin^2(a - b)\right] = \frac{1}{2}, \tag{3}$$

which is **identically** 0.50 **for every possible angle** a.

Thus, **no-signaling is not a probability-space postulate**—it is the Pythagorean trigonometric identity $\cos^2\theta + \sin^2\theta = 1$ of two orthogonal wave projections.

4 Where Does the Nonlocality Reside? (No Propagating Force vs. Shared Wave Frame)

Does this picture eliminate "spooky action at a distance"? It separates two things that are usually conflated:

1. **No Propagating Force or Energy Transfer Between Stations:** In an entangled pair, Station A already holds its own full quantum $\hbar\omega$ and Station B already holds its own full quantum $\hbar\omega$. When Station A locks, **no energy, momentum, or force beam travels through space from A to B**, and Station B's total energy ($\cos^2(a-b) + \sin^2(a-b) = 1$) never changes.

2. **Nonlocality Resides in the Shared Phase Frame of the Wave:** However, the two waves cannot be treated as completely separate objects after leaving the source. They remain a **single, phase-connected wave excitation** ($\phi_A - \phi_B = 0$) spanning the distance between A and B. Locking the phase reference at one end pins the shared phase frame of that extended wave object.

5 Mathematical Proof: Why the First Lock MUST Shift/Entrain the Shared Phase

A natural question is: *Does the first detector's interaction actually have to shift/pin the shared phase, or could the two waves simply leave the source with a shared phase θ_0 that passively guarantees a similar interference pattern at both detectors (outside of the local quantum vacuum and detector phases)?*

Below is the exact proof showing why a **passive (unshifted) source phase θ_0 fails**, and why **entrainment (shifting/pinning the shared phase to the first polarizer's axis a)** is required.

Case A: Passive Source Phase θ_0 (No Shift Upon First Lock)

Suppose the two waves leave the source with a common polarization angle $\theta_0 \in [0, 2\pi)$, and nothing shifts θ_0 when the first detector interacts:

1. At Polarizer A (angle a), the wave interference pattern gives initial shares $s_+(0)_A = \cos^2(a-\theta_0)$ and $s_-(0)_A = \sin^2(a-\theta_0)$.

2. At Polarizer B (angle b), the wave interference pattern gives initial shares $s_+(0)_B = \cos^2(b-\theta_0)$ and $s_-(0)_B = \sin^2(b-\theta_0)$.

3. Each detector then runs its own local $\sqrt{s_+ s_-}\cos\Delta\phi$ walk driven by its own independent detector and quantum vacuum phases. By Theorem 1, for a given source angle θ_0, Station A registers + with frequency $\cos^2(a-\theta_0)$ and Station B registers + with frequency $\cos^2(b-\theta_0)$ independently.

4. Averaging over all source phases $\theta_0 \in [0, 2\pi)$ gives the joint coincidence rate:

$$P_{\text{passive}}(++ \mid a, b) = \frac{1}{2\pi}\int_0^{2\pi} \cos^2(a-\theta_0)\cos^2(b-\theta_0)\, d\theta_0 = \frac{1}{4} + \frac{1}{8}\cos 2(a-b), \qquad (4)$$

which yields the correlation function:

$$E_{\text{passive}}(a,b) = \frac{1}{2}\cos 2(a-b) \implies S_{\text{CHSH}} = \sqrt{2} \approx 1.414. \tag{5}$$

Why Case A Fails Physically (Look at $a = b$): Suppose both polarizers are set to the **exact same vertical angle** ($a = b = 0°$), and a pair leaves the source with $\theta_0 = 45°$. Then Station A gets $\cos^2(45°) = 0.50$ and Station B gets $\cos^2(45°) = 0.50$. Because the local vacuum and detector phases at A and B are independent, in half of those runs Station A's walk goes to $+$ while Station B's walk goes to $-$! Across all θ_0, when $a = b$, the two stations only agree 75% **of the time** ($E(a,a) = 0.50$), whereas experimentally aligned polarizers on $|\Phi^+\rangle$ agree 100% **of the time** ($E(a,a) = 1.00$).

Case B: Active Entrainment / Pinning of the Shared Phase by the First Lock

Now contrast Case A with what happens when the source locks only the **relative phase difference** $\phi_A - \phi_B = 0$, and the first detector to complete its walk **shifts/entrains the common phase orientation into alignment with its winning polarizer axis** (a or $a + 90°$):

1. When Station A locks onto port $+$ (angle a, occurring in 50% of trials), the shared phase is shifted to $\phi_A = \phi_B = a$.

2. Now look at $a = b$: at Station B, the wave arrives with $\phi_B = a = b$, so its Malus interference pattern across Station B's two ports is:

$$s_+(0)_B = \cos^2(a-a) = \cos^2(0) = 1, \qquad s_-(0)_B = \sin^2(0) = 0. \tag{6}$$

3. Because $s_-(0)_B = 0$, the complex interference step at Station B is $2\epsilon\sqrt{s_+ s_-} = 2\epsilon\sqrt{1 \times 0} = 0$. **Even though Station B has its own random detector and vacuum phases, a port with zero wave amplitude cannot fluctuate away from zero.** Station B is therefore guaranteed to register $+$ 100% **of the time** when $a = b$.

4. For general angles (a, b), this gives the exact quantum coincidence rate:

$$\begin{aligned} P_{\text{entrained}}(++ \mid a,b) &= \frac{1}{2}\cos^2(a-b) = \frac{1}{4} + \frac{1}{4}\cos 2(a-b), \\ &\implies E(a,b) = \cos 2(a-b), \qquad S_{\text{CHSH}} = 2\sqrt{2} \approx 2.828. \end{aligned} \tag{7}$$

Table 1: Comparison between a passive source phase θ_0 and shared-phase entrainment upon first lock.

Feature	**Passive Source Phase (θ_0 Unshifted)**	**Shared-Phase Entrainment ($\phi_{\text{shared}} \to a$)**
What the source fixes	Absolute angle $\theta_0 \in [0, 2\pi)$	Relative phase $\phi_A - \phi_B = 0$ (common angle free)
Effect of first detector lock at a	Local only; partner wave stays at θ_0	Entrains shared phase so partner wave is at angle a
Partner station initial shares	$\cos^2(b-\theta_0)$ and $\sin^2(b-\theta_0)$	$\cos^2(a-b)$ and $\sin^2(a-b)$ (Malus's Law)
Same-angle correlation $E(a,a)$	0.50 (75% agreement — **falsified**)	1.00 (100% **agreement — matches experiment)**
General correlation $E(a,b)$	$\frac{1}{2}\cos 2(a-b)$	$\cos 2(a-b)$
CHSH Parameter S	$\sqrt{2} \approx 1.414$	$2\sqrt{2} \approx 2.828$